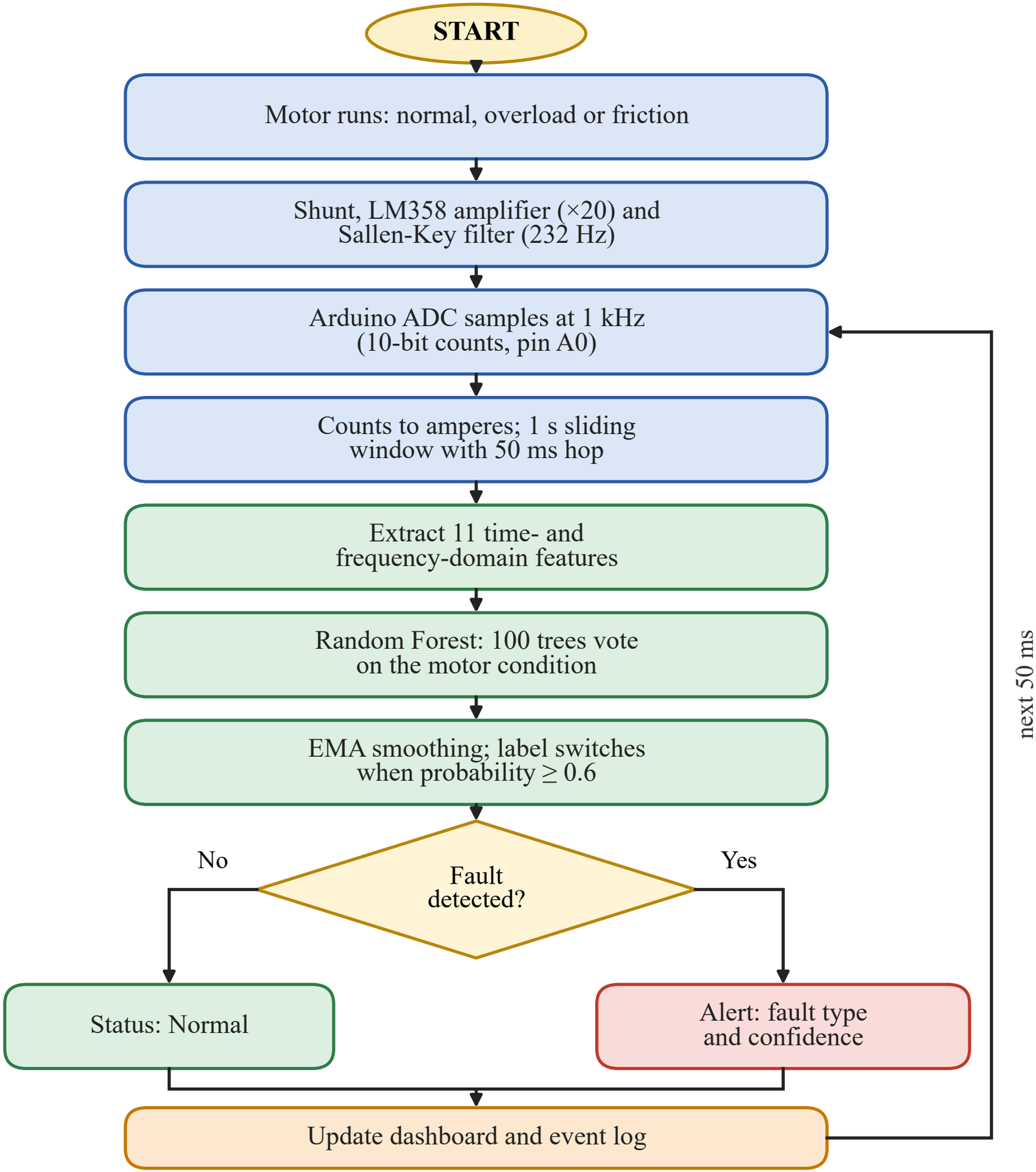
START
Motor runs: normal, overload or friction
Shunt, LM358 amplifier (×20) and Sallen-Key filter (232 Hz)
Arduino ADC samples at 1 kHz (10-bit counts, pin A0)
Counts to amperes; 1 s sliding window with 50 ms hop
Extract 11 time- and frequency-domain features
Random Forest: 100 trees vote on the motor condition
EMA smoothing; label switches when probability ≥ 0.6
Fault detected?
No
Yes
Status: Normal
Alert: fault type and confidence
Update dashboard and event log
next 50 ms